Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.17_2)

Metric

$$ds^2 = -e^{(-2\,\lambda)}\mathrm{d}u \otimes \mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}r - \mathrm{d}r \otimes \mathrm{d}u + r^2 Y\left(u, r\right)^2 \mathrm{d}x_1 \otimes \mathrm{d}x_1 + r^2 Y\left(u, r\right)^2 \sinh\left(x_1\right)^2 \mathrm{d}x_2 \otimes \mathrm{d}x_2$$

Coordinates

$$(x^\mu) = (u, r, x_1, x_2)$$

Metric Functions

$$\Sigma = \sinh\left(x_1\right)$$

$$2H = e^{(-2\,\lambda)}$$

Null Tetrad

Covariant components

$$l = (1)\ du$$

$$n = \left(\frac{1}{2}\,e^{(-2\,\lambda)}\right)\ du + (1)\ dr$$

$$m = \left(\frac{1}{2}\,\sqrt{2} r Y\left(u, r\right)\right)\ dx_1 + \left(\frac{1}{2} i\,\sqrt{2} r Y\left(u, r\right) \sinh\left(x_1\right)\right)\ dx_2$$

$$\overline{m} = \left(\frac{1}{2}\,\sqrt{2} r Y\left(u, r\right)\right)\ dx_1 + \left(-\frac{1}{2} i\,\sqrt{2} r Y\left(u, r\right) \sinh\left(x_1\right)\right)\ dx_2$$

Contravariant components

$$l = (-1)\ dr$$

$$n = (-1)\ du + \left(\frac{1}{2}\, e^{(-2\,\lambda)}\right)\ dr$$

$$m = \left(\frac{\sqrt{2}}{2\,r Y\,(u, r)}\right)\ dx_1 + \left(\frac{i\,\sqrt{2}}{2\,r Y\,(u, r)\sinh\,(x_1)}\right)\ dx_2$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,r Y\,(u, r)}\right)\ dx_1 + \left(-\frac{i\,\sqrt{2}}{2\,r Y\,(u, r)\sinh\,(x_1)}\right)\ dx_2$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{1}{2}\, e^{(-2\,\lambda)}\,\partial_r$$

$$\delta = \frac{\sqrt{2}}{2\,r Y\,(u, r)}\,\partial_{x_1} + \frac{i\,\sqrt{2}}{2\,r Y\,(u, r)\sinh\,(x_1)}\,\partial_{x_2}$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,r Y\,(u, r)}\,\partial_{x_1} - \frac{i\,\sqrt{2}}{2\,r Y\,(u, r)\sinh\,(x_1)}\,\partial_{x_2}$$

Spin Coefficients

$$\rho = \frac{\frac{\partial}{\partial r} Y\,(u, r)}{Y\,(u, r)} + \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{e^{(-2\,\lambda)} \frac{\partial}{\partial r} Y(u,r)}{2\,Y(u,r)} + \frac{e^{(-2\,\lambda)}}{2\,r} - \frac{\frac{\partial}{\partial u} Y(u,r)}{Y(u,r)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cosh(x_1)}{4\,rY(u,r)\sinh(x_1)}$$

$$\beta = \frac{\sqrt{2}\cosh(x_1)}{4\,rY(u,r)\sinh(x_1)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = -\frac{\frac{\partial^2}{(\partial r)^2} Y(u,r)}{Y(u,r)} - \frac{2\,\frac{\partial}{\partial r} Y(u,r)}{rY(u,r)}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{e^{(-2\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)^{2}}{4\,Y\left(u,r\right)^{2}} - \frac{e^{(-2\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)}{2\,rY\left(u,r\right)} + \frac{\frac{\partial}{\partial u}Y\left(u,r\right)\frac{\partial}{\partial r}Y\left(u,r\right)}{2\,Y\left(u,r\right)^{2}} - \frac{e^{(-2\,\lambda)}}{4\,r^{2}} + \frac{\frac{\partial}{\partial u}Y\left(u,r\right)}{2\,rY\left(u,r\right)} - \frac{1}{4\,r^{2}Y\left(u,r\right)^{2}}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{e^{(-2\,\lambda)}\frac{\partial^{2}}{\partial u\partial r}Y\left(u,r\right)}{Y\left(u,r\right)} - \frac{e^{(-4\,\lambda)}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(u,r\right)}{4\,Y\left(u,r\right)} + \frac{e^{(-2\,\lambda)}\frac{\partial}{\partial u}Y\left(u,r\right)}{rY\left(u,r\right)} - \frac{\frac{\partial^{2}}{(\partial u)^{2}}Y\left(u,r\right)}{Y\left(u,r\right)} - \frac{e^{(-4\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)}{2\,rY\left(u,r\right)}$$

$$\Lambda = -\frac{e^{(-2\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)^{2}}{12\,Y\left(u,r\right)^{2}} - \frac{e^{(-2\,\lambda)}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(u,r\right)}{6\,Y\left(u,r\right)} + \frac{\frac{\partial^{2}}{\partial u\partial r}Y\left(u,r\right)}{3\,Y\left(u,r\right)} - \frac{e^{(-2\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)}{2\,rY\left(u,r\right)} + \frac{\frac{\partial}{\partial u}Y\left(u,r\right)\frac{\partial}{\partial r}Y\left(u,r\right)}{6\,Y\left(u,r\right)^{2}} - \frac{e^{(-2\,\lambda)}}{12\,r^{2}} + \frac{\frac{\partial}{\partial u}Y\left(u,r\right)}{2\,rY\left(u,r\right)} - \frac{1}{12\,r^{2}Y\left(u,r\right)^{2}}$$

Weyl Tensor Components

$$\Psi_{0} = 0$$

$$\Psi_{1} = 0$$

$$\Psi_{2} = \frac{e^{(-2\,\lambda)}\frac{\partial}{\partial r}Y\left(u,r\right)^{2}}{6\,Y\left(u,r\right)^{2}} - \frac{e^{(-2\,\lambda)}\frac{\partial^{2}}{(\partial r)^{2}}Y\left(u,r\right)}{6\,Y\left(u,r\right)} + \frac{\frac{\partial^{2}}{\partial u\partial r}Y\left(u,r\right)}{3\,Y\left(u,r\right)} - \frac{\frac{\partial}{\partial u}Y\left(u,r\right)\frac{\partial}{\partial r}Y\left(u,r\right)}{3\,Y\left(u,r\right)^{2}} + \frac{e^{(-2\,\lambda)}}{6\,r^{2}} + \frac{1}{6\,r^{2}Y\left(u,r\right)^{2}}$$

$\Psi_3 = 0$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"